ProofMint

Verifiable AI Agent Certificates on Hedera

HTS (NFTs) | HCS (Audit Trail) | Mirror Node (Verification)

Hedera Hello Future Apex 2026 | AI & Agents Track

Diven Rastdus | Solo Builder

THE PROBLEM

AI agents are black boxes. Nobody can verify what they did.

No Proof of Work

An AI agent says it reviewed your code, analyzed your data, or passed your compliance check. But where's the evidence? Logs can be edited. Databases can be tampered with.

No Accountability

When AI makes decisions that affect real systems, there's no immutable record of what was analyzed, what was found, and what was decided. Trust is assumed, not verified.

> *"Did the AI actually review this, or did someone just click approve?"*

THE SOLUTION

Mint an NFT certificate for every AI task

ProofMint creates a permanent, verifiable, on-chain record every time an AI agent completes work.

AI Reviews Code → Publish to HCS → Mint NFT Certificate → Verify via Mirror Node

Immutable Audit

Full task details (hashes, verdict, issues) published to HCS. Cannot be altered after the fact.

NFT Certificate

Compact NFT minted via HTS referencing the HCS message. Transferable, permanent proof.

Independent Verify

Anyone can query Mirror Node to cross-reference NFT metadata against HCS audit trail.

HEDERA INTEGRATION

Three services, structurally essential

SERVICE	ROLE	WHAT HAPPENS
HTS (Token Service)	Certificate Registry	NFT collection created; each task mints a new serial with compact metadata
HCS (Consensus Service)	Audit Trail	Full task details (input hash, output hash, verdict) published as ordered messages
Mirror Node	Verification Layer	REST API queries fetch NFT metadata + HCS messages for independent hash comparison

Remove any one service and the system breaks. Without HTS: no transferable certificates. Without HCS: no audit trail to reference. Without Mirror Node: no independent verification.

ARCHITECTURE

Tech Stack & On-Chain Proof

Stack

- **hiero-sdk-python** - HTS + HCS transactions
- **FastAPI** - Python backend + REST API
- **Next.js 15** - Certificate dashboard
- **SQLite** - Local state (verified against chain)
- **AI Code Reviewer** - Rule-based + Gemini

Confirmed On-Chain (Testnet)

- NFT Collection: **0.0.8114539**
- HCS Topic: **0.0.8114364**
- 6 NFT serials minted
- 10+ HCS audit messages
- All verifiable on HashScan

TokenCreateTransaction | TokenMintTransaction | TopicCreateTransaction | TopicMessageSubmitTransaction | Mirror Node REST

WHY HEDERA

The only network where this is economically viable

$0.001

Per NFT mint on Hedera. Ethereum: $2-50+ per mint.

Native HCS

Ordered message logging without smart contracts. No Solidity needed.

3-5s

Finality. Real-time certificate minting after each AI task.

Per-task NFT minting is a use case that **only works on Hedera**. On any EVM chain, minting an NFT for every AI code review would cost more than the review itself. Hedera's fixed, predictable fees make high-frequency certification practical.

How ProofMint becomes a business

Business Model Canvas

- **Value Prop**: Tamper-proof AI work certificates at $0.001/mint
- **Customers**: AI agent operators, DevOps teams, compliance-regulated industries
- **Revenue**: SaaS API ($49-499/mo tiers by volume) + enterprise on-prem licensing
- **Channels**: Developer docs, GitHub integrations, AI agent framework plugins
- **Cost Structure**: Hedera fees (~$0.001/tx), compute, storage

Go-To-Market Strategy

- **Phase 1 (Now)**: Open-source SDK, free testnet tier. Target: indie AI developers + hackathon community
- **Phase 2 (3-6mo)**: Paid API tiers. Target: AI SaaS companies needing audit trails (compliance, SOC2)
- **Phase 3 (6-12mo)**: Enterprise. Target: financial services, healthcare, legal AI where verification is mandatory

Key Design Decision: NFT metadata kept under 100 bytes (compact HCS pointer). Full data lives on HCS. This makes minting cheap and scalable while keeping the complete audit trail on-chain.

How ProofMint grows Hedera

3 TXN

per AI task: 1 HCS publish + 1 NFT mint + 1 Mirror Node query. High-frequency usage by design.

1 Account

per customer: Every ProofMint user needs a Hedera account. Enterprise = bulk account creation.

24/7 TPS

always-on: AI agents run continuously. Unlike human users, agents generate transactions around the clock.

Projected Network Impact (Conservative)

MILESTONE	MONTHLY ACTIVE ACCOUNTS	TRANSACTIONS/MONTH
10 teams using ProofMint	10	9,000 (30 tasks/day x 3 txn x 30 days x 10)
100 teams	100	90,000
1,000 teams (enterprise)	1,000+	900,000+

AI agents are the highest-frequency Hedera users possible -- they never sleep, never pause, and generate verifiable on-chain activity with every task.

VALIDATION & MARKET

Who needs this and why now

Target Segments

AI Agent Operators

Companies running AI code review, security scanning, compliance checks. Need proof their AI actually did the work. Growing 10x/year as AI agent adoption explodes.

Regulated Industries

Financial services (SOX compliance), healthcare (HIPAA), legal (e-discovery). AI decisions require immutable audit trails -- not optional, legally mandated.

Market Validation

The Problem is Real

73% of enterprises report they cannot verify AI agent outputs (Gartner 2025). SOC2 Type II auditors now specifically ask about AI decision provenance.

Feedback & Next Steps

Plan: launch open-source SDK on Hiero + publish to npm/PyPI. Track adoption via GitHub stars, API signups, and Hedera testnet transaction volume as early traction metrics.

ROADMAP & LINKS

Where ProofMint goes next

Now

MVP live on testnet. Code review agent + NFT certificates + verification dashboard. Open-source SDK.

Q2 2026

Multi-agent support (any AI task type). PyPI/npm SDK packages. Mainnet launch. First paid API tier.

Q3-Q4 2026

Enterprise tier. Certificate marketplace. Integration plugins for LangChain, CrewAI, AutoGen.

Try it yourself

github.com/astraedus/proofmint

proofmint.vercel.app

hashscan.io/testnet/token/0.0.8114539

hashscan.io/testnet/topic/0.0.8114364

Demo: youtu.be/iqDurB3gVp4

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